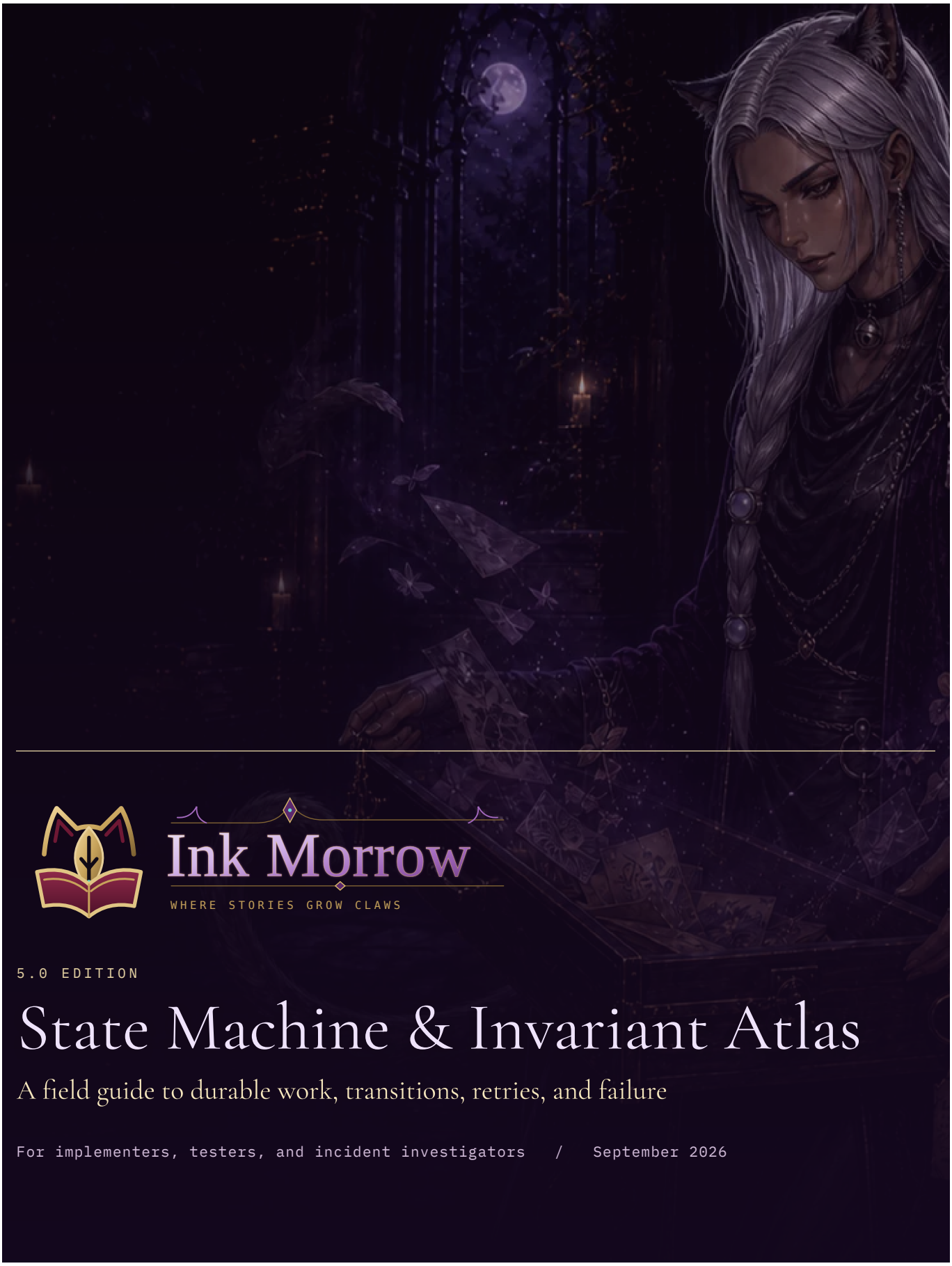


Ink Morrow

WHERE STORIES GROW CLAWS

5.0 EDITION

State Machine & Invariant Atlas

A field guide to durable work, transitions, retries, and failure

For implementers, testers, and incident investigators / September 2026

State Machine & Invariant Atlas

Read each entry as trigger, guard, transition, persistence and failure consequence. A UI success message is not authority: database state and the reviewed purchase boundary are.

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01 Conventions and terminal outcomes

Local means no model request. It can still need the local server and can fail because of access, validation, stale revision or storage. Paid means transport may incur a provider charge even if no scene is accepted.

A request is not a beat. Several quality calls may produce one accepted beat; zero accepted beats may still carry charges. Reader prose, private state, call metadata and provider output are distinct records.

The common successful sequence is intent, expected revision, durable request, reviewed dispatch, validated proposal, stale check and atomic commit. The common failure sequence preserves the old head and records known/unknown spend. A successful replay returns a prior result; it is not another transition.

Snapshot changes are immutable history. Correcting, retiring, changing a preference or taking a role appends state rather than editing past prose. Rewind creates a new path. It is not destructive truncation.

Tests should assert both what changed and what did not: path head, revision, cast ownership, provider call count, private projection and billing. A passed happy path alone cannot prove at-most-once behaviour under races.

B1 adds device navigation states for Desk, Chronicle, Codex, Gallery and Gate, plus a contribution task. Room changes do not activate a different path, advance fiction revision, clear the typed session draft or purchase generation. Opening the explicit active-path selector's command remains a mutation. Reading cursor and actual graph head remain separate; a local record after the last narrative passage is not a new page. B2 also separates read-only view path, inspection boundary and presentation head from the active mutation target. Draft states are opening, local-only, saving, acknowledged, conflict, prepared submission and discarded/expired. Only acknowledgement permits dispatch. A lost response moves to status inspection, never automatic paid retry; a durable cancellation fence resolves an undispatched unknown. A successful action cannot erase newer edits. Lock invalidates reads and queued writes before clearing private UI.

A pending reply permits read-only page/room navigation. Completion updates current metadata and announces new prose; a reader who moved away from the submitted tail stays in place. Task-close is distinct from request cancellation. An unchanged submitted draft may be cleared on success; a newer revision with identical text is preserved. Lock clears every private room/task buffer immediately, before awaiting server revocation.

02 Start, continue and clarify

Trigger	Guard	Durable result
Start own situation	Valid title/premise/cast/preferences	New game/path, no paid prose
Start curated situation	Known scenario and valid additions	Authored opening, cast and initial facts
Follow	Active episode, current revision, available reviewed plan	One accepted narrated beat
Steer	Valid direction and scope	Beat; ongoing focus only when explicitly selected
Ask	Valid outside-story question	Clarification, no fictional scene advancement
Submit raw opening prose	Production route rejects the field	No game created

A pending purchase blocks conflicting local mutations. Invalid input fails before dispatch. If the selected role is unavailable, the game does not silently choose a different model. A draft response must pass structural validation even when quality is Off.

Ask is still a paid response, but cannot invent an in-world event, grant a challenge, increment the fourth-wall scene cooldown or resolve an episode goal. It is not a hidden-solution endpoint.

Neither starting a story nor reading its existing opening starts a provider call. There is no speculative successor waiting behind Continue. Busy feedback names actual work rather than disguising a second background purchase.

03 Character control and local correction

Inhabit validates a current cast identity and appends a handoff snapshot. Act and Say require that explicit control. Release appends a return to reader-director. None purchases text by itself or replaces the story graph.

Generated effects cannot choose an inhabited person's speech, commitments, decisions or inner relationship expectations. A proposed interaction can leave a choice for the player. Structural guards do not prove every prose sentence respects that ownership; model evaluation remains necessary.

Correction validates an existing public fact or a new fact and a reason, then records an authoritative local change. Earlier prose remains intact. Retirement removes a fact from future retrieval on this path while preserving historical evidence. Neither operation is a memory quota maintenance chore.

A correction and an alternate choice are different transitions. Fix a record that is wrong; fork when exploring another possible history. Local changes do not manufacture narrated payoff simply by marking a goal complete.

Reader evidence omits private reasons. A source link must point to a real ancestor, not a future beat or another path. Import remaps the same constraint to fresh IDs.

04 Path creation and selection

Explore an alternative selects an exact current or earlier moment, checks revision and idle state, then creates a named branch at that snapshot. Rewind uses the same mechanism and preserves the original path.

Selecting an existing branch checks its game ownership and current revision, then changes active selection. Facts, knowledge, resources, commitments, relationships, quality, play style, fourth-wall cooldown, focus, control, episode and placements come from that branch's state.

No branch merge is implied. A promise made after the fork on one side is absent from the other. A reader page outside current ancestry cannot be used as evidence merely because it exists elsewhere in the same story.

At most 40 paths are supported. A limit or stale-revision failure creates no partial path and retains the dialog draft. Refresh before choosing again; never recover a failed local fork by buying narration.

A pending paid request prevents a branch change from becoming a hidden target switch. A late response rechecks exact path and revision before commit. UI route fencing independently prevents an old response from painting another screen.

05 Provider assignment, vault and consent

A role assignment names profile and model. Saving it is local and does not validate quality by buying a completion. Catalogue browsing is a provider read. An unavailable saved model remains unavailable until deliberately repaired.

Session credentials exist in process memory. Vault storage encrypts credentials under a data key wrapped from the owner password. Login can unlock; lock, terminal reset and process disposal remove plaintext access. A remembered browser session after restart does not itself decrypt the vault.

Ordinary paid consent is remembered on the device. Expanded quality uses a separate identity derived from the complete role/provider/model plan and call ceiling. Neither is a substitute for auth, CSRF or the server's revision checks.

A changed plan invalidates purchase authority. Missing memory support cannot be silently omitted from Both. A transport failure does not authorise retry, and a quality rejection does not earn a second repair.

Imported saves contain preferences and aggregate spend, not credentials, consent, call IDs or request keys. Import therefore cannot resume a purchase or authorise a new one. Free local repeated rulings remain independent of provider and quality availability.

06 Authentication and database startup

Static boot is gated. Status resolves setup-needed, locked or unlocked. Setup requires a terminal code and valid password. Login creates an opaque session, its digest is stored, and mutations require the session CSRF token. Lock revokes one session; password change revokes the others.

Private API guards run before body parsing. Before unlock, even a large body cannot become an import, upload or generation job. Production auth is explicit in server.js and cannot be disabled by NODE_ENV=test.

Database startup resolves one file, inspects a private scratch copy of existing database/journals, validates family/version/ledger/integrity, and only then opens the original for normal WAL and transactional migration. Failure cleans only private scratch space, not source data.

Missing/empty source plus orphan journals is rejected. Existing old-family, unknown, corrupt or future databases are not adopted. An accepted 5.0 migration is contiguous and transactional; failure leaves the prior valid version.

After startup, abandoned model requests become interrupted. Their already dispatched uncertain calls remain unknown-cost, not free. No background continuity rebuild, old image queue or retired share service starts.

07 UI states and regression traces

UI state	Allowed action	Required protection
Loading local data	Wait, leave view	Immediate status; late-result fencing
Editing a draft	Edit, cancel, choose input kind	No purchase while typing
Reviewing purchase	Confirm or cancel	Duplicate review guard; retain draft
Submitting	Read busy/progress state	No second paid POST
Failed	Inspect error, free refresh	No automatic paid retry
Locked	Unlock only	Clear private DOM/state and stale callbacks
Ended episode	Read or begin another	No forced continuation

For any network mutation, delay its response, navigate elsewhere, then resolve it. Assert both server accounting and absence of stale UI mutation. Repeat with Lock and a second tab's revision change. Test cancellation before dispatch and failure after a known charge separately.

For uploads/imports, inject validation and transaction failures and inspect staged-file cleanup. For paths and evidence, test sibling/future references. For every optional quality plan, enumerate its maximum calls and the one-repair budget. A passing reviewer must not bypass deterministic state guards.

Use the real production composer without legacyEnabled for boundary tests. Old module fixtures may opt into the retired seam explicitly; their passing tests do not demonstrate that the writing suite is exposed in 5.0.

08 Optional quality transitions

Off: dispatch draft → validate → commit or fail. Enabled: dispatch draft → validate → selected review(s) → commit if approved. An evidenced rejection permits one repair → validate → all selected reviews → commit or fail. An invalid initial draft can consume the same sole repair before review. Invalid reviewer output, transport failure, stale state or changed provider configuration terminates work; none grants another call. One reviewer caps at four total calls, both at six.

Each call moves pending → completed or failed; restart moves pending → interrupted. A late completion may replace interrupted billing uncertainty but cannot reactivate the parent request or append a scene. Dispatched unknown costs are not zero. Known charges remain counted when another call lacks a price. A parent with call rows is excluded from the parent-only spend sum. Exactly one accepted beat and state snapshot commit, regardless of how many draft/review calls were bought.

An unchanged challenge ruling takes its local zero-call path before provider or quality availability checks. Successful idempotency replay also buys nothing. Quality consent is scoped to the server plan; old one-call consent and saved story data cannot authorise an expanded purchase.

09 People and episode transitions

Relationship development: quoted passage/input evidence → existing relationship identity → ownership check → new description with unchanged aspect, people and visibility → immutable change with prior provenance. World-fact replacement is not a develop transition. Affection does not imply trust or cooperation.

Episode opening → public developments → narrated resolution of its recorded goals → payoff → subsequent aftermath. These are descriptive opportunities, not forced story beats. No transition ends the episode: the player explicitly ends it or starts the next one. Quiet turns and local goal corrections do not fabricate a payoff. Rewind restores the earlier snapshot; saves remap payoff evidence and reject missing, cross-path or non-scene payoff references.

Catch me up → public current-path read → bounded existing summaries and records. There is no purchased recap, autonomous absence step or simulated offline progress. Failed path/episode writes retain their local dialog drafts and create no new path.

10 Fourth-wall transitions

Local preference change → snapshot the Never/Rarely/Freely choice, leaving the last-address index intact. Narration → compute permission from play style, participation, eligible non-inhabited cast and scene gap → validate optional aside → append named address and updated index with the complete scene. Rarely permits at most one structured address in six narrated scenes. Failed, stale, Ask and free repeat operations cannot advance the cooldown. No timer runs while away.

A forbidden address fails the response without partial prose or state; actual provider spend remains recorded. Rewind restores the earlier index; save import validates it against scene count and preserves it in the new copy. An aside is neither world-state evidence nor a new source of challenge authority.

11 Influence transitions

Invitation selection -> editable direction only, no request. Moment Steer -> one response, unchanged ongoing focus. Ongoing Steer -> validated response and focus change in one commit. Failure/cancel -> retained draft and scope, unchanged focus. Clear focus -> explicit local preference snapshot. Rewind restores the prior focus and style. Acting, speaking and asking cannot set an ongoing editorial focus.

Challenge review -> local comparison at expected revision -> either ordinary paid consent or reuse of the recorded ruling. A changed revision fails before dispatch. Source navigation -> current-ancestry check -> reader-safe beat, never a private state snapshot. Prior evidence references cannot point to a future or sibling path.

12 Resistance and memory transitions

Explicit approach: validate intent and challenge -> resolve requirements against current-ancestry facts -> determine outcome and stable basis. An unchanged prior decision -> local clarification, zero provider attempts, unchanged fictional scene count. New grounds -> one reviewed narration -> check returned outcome/evidence -> atomic prose, state, adjudication and usage. Wrong structured outcomes fail without a partial commit; known charges remain. Grants remain recorded after later changes to the grounds; rewind restores the previous decision set rather than rerolling it.

Facts: append evidence -> keep immutable change -> retain a bounded working set. Historical retrieval chooses the latest fact version on the active ancestry. Correction supersedes; retirement hides from retrieval; neither deletes history. Forks cannot retrieve a sibling's future. Saves preserve challenge definitions and adjudication evidence, validate ancestry and remap moment identities on import.

13 Illustrated-path and save transitions

Local upload: validate target -> normalize raster -> stage private file -> append placement snapshot and asset metadata atomically. Failure before commit removes only the staged file. Describe/remove appends another snapshot; no prose is edited. Fork/rewind restores the exact placement set at that moment. Historical assets stay.

Paid painting: journal pending -> check reviewed Illustrator -> mark dispatched -> one provider attempt -> validate raster -> compare revision/path -> atomically save asset, placement and terminal usage. Failure/interruption retains known or unknown spend and never retries. Completed keys replay for free even after path changes.

Save: idle story -> complete graph/media snapshot -> bounded gzip file. Import: bounded decompression -> strict graph/media validation -> read-only preview -> explicit copy -> stage new media -> single deferred-FK transaction -> new story. Rollback removes its staged files, never original data. Imported aggregate spend is terminal; request keys, credentials and consent do not travel. Books are a separate read-only active-path projection, with EPUB image-only spine pages before text.

Catalogue creation/editing: validate typed fields -> compare expected revision -> require idle entry -> save metadata and increment revision. No provider call. Upload follows raster normalization and stale recheck; replacement cleans only the previous catalogue image after commit. Catalogue deletion scrubs content and removes its owned image but preserves image accounting. Frozen story copies are not targets.

AI reference development: collect bounded seed fields -> review Storyteller, model, private-field exposure and two-call ceiling -> journal pending by idempotency key -> dispatch once -> validate the complete typed result -> if invalid, dispatch one correction -> return an editable unsaved form. Cancel saves nothing. Failure or restart records known/unknown spend and requires a new explicit take; successful key replay returns the same draft without purchase. Saving and painting remain separate.

Catalogue painting uses its own pending/succeeded/failed/interrupted journal, one pending request per entry and one dispatch per key. A changed provider or revision refuses attachment. Restart marks pending work interrupted; a late completion can settle its known cost but cannot make the image current. There is no silent retry.

Setup selection freezes world, Scribe and character references, copies normalized images to fresh story-owned identities and commits the graph atomically. Story cover and reference updates append path-local snapshots through the story request journal. Saves validate/remap every visual asset reference. Covers enter books; other reference art remains private. Images do not create facts, knowledge or cast control.